DEWDRIVE · CALCULATION

# DewDrive sizing calculations

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## DOCUMENT CONTROL

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## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.9 | 2026-10-25 | Amish Chadha | Amish's decisions of 2026-10-25: design wind a 25 m/s gust (item 4): 495 N on the lid, 234 N per anchor or 96 kg of ballast; fan cut-out 65 % RH, restart below 60 % (item 3) |
| 0.8 | 2026-10-25 | Amish Chadha | STEM check fixes and independent review (DWD-STM-001; DWD-DEC-001 D16 to D20): black perforated absorber plates on the beds (C11), open tray floor of mesh on bars (vapour open share 0.59, was taken as 0.60 for a floor that passes about 0.25), sensitivities C12 and C13, peak-day stagnation D6, gust check H5, PTFE coatings and painted condenser underside in the BOM; requirement IDs moved to DWD-PRS-001; tagged values written to sizing_out.json; PR-001 not met (0.44 L per day); cost USD 695 against the USD 520 target |
| 0.7 | 2026-10-02 | Amish Chadha | Script re-run with R9 as restated (computed counts now agree); paper study of a passive stagnation vent (D4, D5); BOM line 3 repriced for a 130 °C |

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| | | | sheet and label added; cost USD 639 against the USD 520 target |
| 0.6 | 2026-10-02 | Amish Chadha | R9 row and counts follow the restated R9; glazing rating decision noted in section D (DWD-DEC-001, items 2 and 3); no computed number changed |
| 0.5 | 2026-10-01 | Amish Chadha | Budget treated as a value-engineering target |
| 0.4 | 2026-10-01 | Amish Chadha | Re-run on the constructable model (DWD-DDR-003); mass by component and material, outlet slot in the fan duty, new leg span in the wind check, repriced BOM; R9 and R11 not met |
| 0.3 | 2026-09-26 | Amish Chadha | Budget top-up approved by Amish (budget $520; R11 met); script re-run |
| 0.2 | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002); design of record now through-flow with 25 wt % salt, drip screens and fan cut-out; paper evaluation of a low-emissivity screen |
| 0.1 | 2026-09-25 | Amish Chadha | First issue for TRL 3 (sorbent isotherm, night adsorption, day desorption and condenser, cyclic yield, stagnation and touch temperatures, salt containment, electrical energy, mass, wind, logging, cost) |

# DewDrive sizing calculations

This revision checks the design decided by Amish on 2026-09-25 (DWD-DDR-002): night air drawn down through the sealed, mesh-floored trays; 25 wt % $CaCl_2$ (1.0 kg in 3.0 kg of gel); a black drip screen under each tray; and a fan that stops above 70 % RH. v0.8 adds the fixes of the first STEM check and of its independent review (DWD-STM-001; DWD-DEC-001, D16 to D20). Up to v0.7 this note took the sun to be absorbed at 95 % "on the black tray tops", but 93 % of the tray footprint is the sorbent bed, whose beads are white; with the beads' own absorptance (about 0.3) the design would make only 0.01 L per day [C11]. A black perforated aluminium plate now rests on each bed, with an effective absorptance of 0.89 [C11]. The independent review then found that v0.7 let the vapour through 60 % of the tray floor, where a 50 % perforated sheet under a fine mesh passes only about 25 %; the floor is now a coarser mesh on thin bars, 59 % open (D19). On paper the design, as made constructable in DWD-DDR-003 and fixed in v0.8, meets nine of the thirteen requirements it shares with DWD-REQ-001 (PR-001 to PR-013 of DWD-PRS-001). **It collects about 0.44 L per day at 40 % night RH, below the 0.5 L target of PR-001**, and 0.31 L at 25 % (PR-002 target 0.25 L), against 0.20 and 0.09 L for the v0.1 layout. The calculation's own sensitivities put it lower still: 0.41 L per day with the temperature drop through the bed [C12], 0.38 L if grain kinetics cut the night uptake [C9] and 0.37 L for the pessimistic combination [C13]. The salt stays in the pores after three 90 % RH nights in a row. Making the design buildable added the parts that hold it together (wall battens, a tray deck, screen frames, stand bracing and fixings), and the STEM check fixes added the absorbers, PTFE coatings, floor bars and paint: the estimated parts cost is now $695 against the $520 value-engineering target (PR-011 over the target by $175) and the full box weighs 36.1 kg, 23.4 kg once the trays and deck are lifted out (R9 was not met as first written; Amish restated it on 2026-10-02 for a lift with the trays and deck out, and PR-009 is met). The design decisions register (DWD-DEC-001) lists cost drivers and savings for R11, and the ways to recover PR-001 for Amish to decide. A low-emissivity screen, evaluated here as decided, would lift the yield to 0.82 L per day but let a dry bed stagnate at about 159 °C, beyond the usual rating of polycarbonate glazing, so it is not adopted; the paper study of a passive vent in section D finds no passive remedy and the screens are dropped. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [B2], is the line of that script's output that carries it.

> **Safety:** These calculations concern hot surfaces (a dry bed up to about 133 °C on a peak day), a corrosive calcium chloride solution, a lithium iron phosphate battery and a 1.1 m² panel in wind. They are first-principles estimates for a paper proof of concept, not a substitute for measured isotherms, supplier data, an engineering review or test. Harvested water must be tested and treated before drinking. See DWD-PRC-001, Safety.

## Scope and method

The note checks every requirement in DWD-PRS-001 v1.0 against the design in DWD-PRC-001 v1.0 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `derived()` and components (`build_components()`, each with its material), so the areas, gaps and masses used here are those of the STEP files and of drawing DWD-DWG-001 Rev P6. It reads prices from `bom/bom.csv`, writes the requirement table to `docs/04-calcs/results.csv` and the tagged values to `docs/04-calcs/sizing_out.json`, which the STEM check script reads. Run it from the repo root with `python docs/04-calcs/sizing.py` (about 8 min; the yield is a time-stepped simulation of repeated night and day cycles until the bed water at dawn repeats).

"Design" means the design of record after DWD-DDR-002. "v0.1 layout" is the earlier design, in which the night air flowed past the trays at 32.5 wt % salt; its figures are kept for comparison and come from v0.1 of this note unless tagged.

## Assumptions

*Table 1. Main assumptions.*

| Area | Assumption | Basis |
|---|---|---|
| Design point | Night 10 h at 20 °C and 40 % RH (low-humidity case 25 %); day air 35 °C; 6.0 kWh/m² per day on the tilted aperture, as a sine from 06:00 to 18:00; flaps shut at 07:00 and opened at 19:00, bed back at air temperature when the fan starts | DWD-REQ-001 |
| Psychrometrics | IAPWS-IF97 saturation pressure; vapour diffusivity 2.5 x $10^{-5}$ m²/s at 20 °C, scaled with T^1.75 | Standard |
| Salt solution | Water activity of aqueous $CaCl_2$ from the Conde (2004) correlation; solubility 74.5 g/100 g water at 20 °C, 81.1 at 25 °C, 134.5 at 60 °C and 152.4 at 100 °C | Conde, *Int. J. Thermal Sciences* 43, 2004; Wikipedia |
| Salt hydrates | Below deliquescence at night: hexahydrate above 18.5 % RH, tetrahydrate from 9 % to 18.5 %; above 29.9 °C the solid is taken as the dihydrate | SaltWiki, calcium chloride (transitions at 20 °C) |
| Composite | 3.0 kg mesoporous silica gel, 1.0 cm³/g pore volume, skeleton 2,200 kg/m³, packing 0.60; 1.0 kg $CaCl_2$ (25 wt %, DWD-DDR-002 D10); the gel walls add 0.08 kg/kg per unit water activity; no pore confinement effect on the salt | Typical wide-pore gel; engineering judgment |
| Heat of sorption | Clausius-Clapeyron on the isotherm at constant bed water, not less than the latent heat | First principles |
| Night mass transfer, design | Air drawn down through the bed: packed-bed Wakao correlation for 3.5 mm beads; diffusion inside the grains not modelled (see C9) | Standard; grain kinetics must be measured |
| Night mass transfer, v0.1 layout | Air splits between the 45 mm gap over the trays and the 65 mm gap under them in proportion to gap cubed (laminar); Nu 4.86 plus a developing-flow term; Lewis analogy; diffusion into the bed with porosity 0.40 and tortuosity 1.5; mesh floor 60 % open | Standard laminar-channel results |
| Fan cut-out | Fan stops above 65 % RH and restarts below 60 % (lowered from 70 % by Amish on 2026-10-25, DWD-DEC-001 item 3); natural airflow through the open flaps then taken as 4 m³/h, 10 % of the fan flow | Assumption; to check by test |
| Day, bed | Lumped bed: sorbent, four 0.6 kg aluminium trays, the four absorber plates (1.71 kg) and the water; transmittance 0.78 times an effective absorptance of 0.89 for the black plate (22.7 % open; 0.95 on the plate, 0.70 for light through the holes, an estimate), so 0.70 over the tray footprint; loss to ambient 3.2 W/(m²·K) through the gap and twin-wall glazing. v0.7 took 0.74 on the white beads, which absorb about 0.3 | Twin-wall polycarbonate U value about 3 W/(m²·K); DWD-DEC-001 D16 |
| Day, vapour path | Stefan diffusion through still air from the bed underside across the 73 mm gap to the condenser (hot above cold is stably stratified); 59 % of the bed area open through the floor (PTFE-coated glass-fibre mesh about 60 % open on bars 97.5 % open, D19; v0.7 took 60 % for a perforated sheet under a fine mesh, about 25 % in fact). The drip screen (two 18 mm deep layers of channels, one third open) adds 18 mm x (1.5 / 0.33 − 1), giving an equivalent path of 136 mm | First principles; geometry from the model; independent review |
| Day, parasitic heat | Radiation from bed to condenser: mesh and beads (emissivity 0.9) seen between the floor bars, bare aluminium 0.1 on the bars, plate 0.9; the drip screen blocks every line of sight and acts as a radiation shield with emissivity 0.90 on both faces (black PTFE), or 0.15 for the low-emissivity option; conduction across the gap; sun through the gaps between trays onto the plate, absorptance 0.5 | Engineering judgment |

| Area | Assumption | Basis |
|---|---|---|
| Condenser | 2 mm aluminium plate and 21 fins, 1 x 100 x 920 mm; laminar natural convection on vertical fins; radiation from the fin envelope (emissivity 0.85, fins and plate underside painted, DWD-DEC-001 D18) to shaded ground at ambient; aluminium 205 W/(m·K) | Model; standard correlations |
| Collection | 30 g per day left as drops and films on the plate and gutter | Engineering judgment |
| Electrical | Fan 2 W for 10 h; logger 0.15 W average; 10 W panel, 5 peak sun hours, 0.70 derating; 12.8 V 6 Ah battery, 80 % usable | DWD-PRC-001 |
| Wind | 25 m/s gust until the site value is known (DWD-DEC-001 item 4, decided by Amish 2026-10-25; was 20 m/s); force coefficient 1.2 on 1.1 m² normal to the lid; centre of pressure a quarter chord upwind of the centre; friction 0.5; factor of safety 1.5 | Engineering judgment |

## A. Air and sorbent

At 20 °C the night air carries 6.92 g/m³ of vapour at 40 % RH and 4.32 g/m³ at 25 % RH [A1]. The Conde correlation puts the deliquescence point of $CaCl_2$ at 20 °C at a water activity of 0.341, against about 0.30 quoted by SaltWiki [A2]; the difference only matters between 30 and 34 % RH. At 40 % RH the salt is a 40.1 wt % solution holding 1.49 kg of water per kg of salt; at 25 % RH it is the hexahydrate, holding 0.974 kg/kg [A3].

With 1.0 kg of salt the whole bed would hold 1.59 kg of water at equilibrium at 40 % RH (0.397 g/g) and 1.03 kg at 25 % RH (0.258 g/g) [A4], less than the 2.03 and 1.32 kg of the v0.1 loading but still well above what a night delivers. The 3.0 kg of gel has 3.00 L of pore volume; the bed is 7.27 L of beads, 9.3 mm deep on 0.785 m² of tray floor, which matches the model [A5].

## B. Night adsorption

The fan moves 40 m³/h, so 2.77 kg of vapour passes through the box in a night at 40 % RH, and 1.73 kg at 25 % RH [B4].

**Design: air drawn through the bed.** The superficial velocity through the bed is 14.2 mm/s and the gas-side number of transfer units is 19, so the air leaves close to equilibrium with the bed; the bed adds only 0.16 Pa of pressure drop [B3]. In the repeating cycle the bed takes up 0.47 kg a night at 40 % RH (a swing of 0.117 g/g), from 0.81 kg at dusk to 1.27 kg at dawn, still below the 1.59 kg equilibrium; the heat of adsorption warms it to 24.7 °C [B5]. At 25 % RH it takes up 0.34 kg [B6]. The night is not the limit: the uptake is set by how dry the day leaves the bed.

**v0.1 layout, for reference.** Air flowing past the trays splits 25 % over and 75 % under them, stays laminar, and reaches NTU 0.31 and 0.06, capturing at most 11 % of the vapour driving force [B1, B2]. That is why v0.1 took up only 0.23 kg a night.

## C. Day desorption, condenser and yield

The aperture receives 6.00 kWh/m² per day, of which 4.24 kWh/m² falls between 09:00 and 15:00; the peak is 785 W/m² [C1].

Table 2 traces the design point. The bed passes 83 °C by 09:00 and reaches 105 °C at noon [C2]. Vapour moves from about 08:30 to 16:00, peaking at about 96 g/h. By late afternoon the salt has dried towards the dihydrate and desorption stops.

*Table 2. Day at the design point (script output, section C).*

| Hour | Sun (W/m²) | Bed (°C) | Condenser (°C) | Bed water activity | Desorption (g/h) | Bed water (kg) |
|---|---|---|---|---|---|---|
| 07:00 | 203 | 24.8 | 34.0 | 0.312 | 0 | 1.27 |
| 09:00 | 555 | 82.8 | 42.3 | 0.371 | 32 | 1.26 |
| 11:00 | 759 | 101.8 | 46.3 | 0.347 | 92 | 1.13 |
| 12:00 | 785 | 105.3 | 46.9 | 0.314 | 95 | 1.03 |
| 13:00 | 759 | 105.8 | 46.6 | 0.279 | 81 | 0.94 |
| 15:00 | 555 | 95.7 | 43.9 | 0.223 | 31 | 0.83 |
| 17:00 | 203 | 68.4 | 39.4 | 0.183 | 0 | 0.81 |

*Table 3. Yield and energy.*

| Quantity | Design (black screens) | No screens | Low-emissivity screens | v0.1 layout | Tag |
|---|---|---|---|---|---|
| Collected at 40 % night RH | 0.44 L per day | 0.31 L per day | 0.82 L per day | 0.20 L per day | C3, C6, C10 |
| Collected at 25 % night RH | 0.31 L per day | | 0.54 L per day | 0.09 L per day | C4, C10 |
| Bed at noon; peak | 105 °C; 106 °C | peak 91 °C | 127 °C; 130 °C | 104 °C | C2, C6, C10 |
| Condenser rise above ambient, peak | 11.9 K | 13.3 K | 9.6 K | 13.1 K | C2, C6, C10 |
| Condenser load, peak | 292 W | 334 W | 226 W | 328 W | C2, C6, C10 |
| Stagnation, dry bed | 111 °C | 90 °C | 159 °C | 105 °C | D1 |
| Solar-to-water efficiency | 5.4 % | | 10.1 % | 2.5 % | C7 |

The design, no-screen and low-emissivity columns include the absorber plates and the open floor of v0.8; the v0.1 column is kept from v0.1 of this note.

The "no screens" column is the through-flow design at 25 wt % without drip screens. The effective emissivity from the bed underside to the condenser is 0.802 with no screen, 0.405 with black screens and 0.074 with low-emissivity screens (the open floor of D19 shows more of the emissive mesh and beads than v0.7's perforated sheet did); the screens nearly double the equivalent vapour path, from 73 to 136 mm [C5].

The energy balance shows where the sun goes in the design at 40 % RH [C7]: 5.54 kWh falls on the inner aperture; the absorber plates and tray rims absorb 3.46 kWh; 0.33 kWh drives desorption; 1.53 kWh is lost back through the glazing; and 1.52 kWh passes from the hot bed to the condenser (0.61 kWh with low-emissivity screens).

The condenser has 3.86 m² of fins with a natural-convection coefficient of 4.7 W/(m²·K) and a fin efficiency of 0.87, giving 24.6 W/K at a 12 K rise [C8], of which radiation from the painted underside (D18) is about a third. It holds the peak rise to 11.9 K, inside PR-005 (R5). The independent review points out that the correlation is for an isolated vertical plate, while these fins hang under a hot, tilted floor that traps warm air: 30 % less convection would give about 14 K, and half the radiation as well about 16 K, so PR-005 is at risk until the condenser is logged at TRL 4.

**Item 5 evaluation (DWD-DDR-002, D12).** A radiation screen under the trays does what the TRL 3 review expected: the less heat the bed sheds to the condenser, the hotter and drier it gets by day, and the cooler the condenser stays. The black drip screens needed for D11 already give part of this effect (0.31 to 0.44 L per day, even though they double the vapour path). A low-emissivity screen gives much more (0.82 L per day, 10.1 % efficiency), but with an empty or dry bed the trays would stagnate near 159 °C, well above the roughly 120 °C service rating usual for twin-wall polycarbonate. Adopting it would need a stagnation vent or a glazing rated for that temperature. That is proposed as a new item (O4), not adopted.

**Sensitivity.** Diffusion inside the grains is not modelled. If it cut the effective night NTU from 19 to 2 or to 1, the design would collect 0.42 or 0.38 L per day [C9].

**Absorber (STEM check S5, DWD-DEC-001 D16).** The sun reaches the beds through 10 mm twin-wall polycarbonate (transmittance about 0.78). Up to v0.7 this note took an absorptance of 0.95 on the "black tray tops", but the trays are open pans and the sun falls on the sorbent, whose silica gel and calcium chloride beads are white; with an absorptance of about 0.3 (an estimate) the design collects only 0.01 L per day and the bed reaches 64 °C at noon [C11]. Each bed now carries a black perforated aluminium plate, 1 mm thick with 3 mm holes at 6 mm staggered pitch (22.7 % open), black PTFE-coated. The plate absorbs 0.95 of the light on its solid part; of the light through the holes, the beads absorb some and scatter most back onto the black underside of the plate, taken together as 0.70 (an estimate to measure at TRL 4; the independent review notes that 3.5 mm beads sit right under 3 mm holes, so more may escape). The effective absorptance is 0.89 and the transmittance-absorptance 0.70; the four plates weigh 1.71 kg and their heat capacity is in the lumped bed [C11]. A 35 % open plate was tried first and gave less water, so the closer pattern was chosen.

**Temperature drop through the bed (C12) and the combined case (C13).** With the heat entering at the plate on top, conduction through the 9.3 mm bed (about 0.2 W/(m·K)) gives about a 16 K drop from top to bottom at noon. Desorbing at the mid-depth temperature, half that drop below the plate, the design collects 0.41 L per day [C12]. Taking that together with half, not 0.70, of the light through the absorber holes absorbed gives 0.37 L per day [C13]. The lumped model, used for every other figure, is the optimistic end.

**Tray floor (independent review; DWD-DEC-001 D19).** v0.7 took 60 % of the bed area as open to the vapour through the tray floor, but a 50 % perforated sheet under a fine mesh passes only about 25 %: with that correction the design would collect 0.36 L per day. A 70 % open expanded floor under a 1.5 mm mesh gives 0.41 L per day. The floor is now a PTFE-coated glass-fibre mesh of about 1.8 mm aperture (about 60 % open, an estimate) on seven aluminium bars 6 x 1.5 mm on edge (97.5 % open), 59 % open in all; it gives 0.44 L per day. More of the emissive mesh is then seen from the condenser, which costs part of the gain. The floor bars raise the bed 6 mm in the tray, so the bed underside is 73 mm above the plate.

## D. Stagnation and touch temperatures

With a dry bed at noon on a 35 °C day, the bed stagnates near 111 °C with black drip screens, 90 °C with no screens and 159 °C with low-emissivity screens [D1]. The independent review asked for a peak day: at 1000 W/m² on the aperture, 40 °C air and the condenser 12 K above the air, a dry bed reaches about 133 °C and the glazing inner skin, which is what the sheet's rating applies to, about 106 °C (88 °C at the design point) [D6]. Amish decided on

2026-10-02 (DWD-DEC-001, item 3) to specify a sheet rated 130 °C or more, or a 120 °C sheet only if the supplier's data and the inner-skin temperature measured at the TRL 4 stagnation test both show it stays inside its rating. The outer skin of the glazing reaches about 55 °C at stagnation [D2], and the condenser fins, at ankle and hand height, up to 47 °C [D3]. Both are under the 60 °C limit of PR-013 (R13), but the inside of the box is not: the bed and trays are hot enough to burn. The plywood, PIR, EPDM and silicone near the hot interior see gap air of about 85 °C on a peak day; their ratings are to be confirmed with suppliers (DWD-STM-001, S27).

*Passive stagnation vent study (decided on 2026-10-02, DWD-DEC-001, item 4).* The figures in this paragraph are those of v0.7, when the study was made and concluded; rerun with the absorber plates and open floor of v0.8 the script gives 159 °C, a vent of about 98 cm² and a peak of 130 °C, which leaves the conclusion unchanged [D4, D5]. The low-emissivity screens would raise the dry-bed stagnation temperature to 166 °C. A vent at the high north edge of the box, with its inlet low on the south side 100 mm below, would let hot gap air leave by stack effect. Taking the gap air at the mean of bed and ambient temperature, a discharge coefficient of 0.6 and the same heat balance as D1, holding the dry bed at 122 °C needs about 117 cm² of opening, about a 15 mm tall slot along the 800 mm inlet width; the same vent would bring the black screens to 103 °C [D4]. The vent would have to stay shut in normal operation, because with low-emissivity screens the bed already reaches 130 °C at noon and 149 °C at its peak, above the 122 °C it must hold, and any open vent would let water vapour escape that the condenser should collect [D5]. A passive flap therefore cannot do the job; it would need a thermostat that tells a dry bed from a working one, which is an active part with its own cost (a wax actuator, flap and seal, about $20 as an estimate) on top of a design that is already over its cost target, and the benefit has not been shown beyond the paper gain in yield (0.82 L per day against 0.57 L per day at 40 % RH, C10). The study therefore concludes that the vent cannot hold a dry bed near 122 °C passively at a modest cost, and the low-emissivity screens are dropped, as the 2026-10-02 decision provides. The first prototype keeps the black screens.

## E. Salt containment

A calcium chloride solution stays in the beads only while its volume is less than the pore volume. At equilibrium at 40 % RH the solution fills 58 % of the 3.00 L of pores; at 90 % RH it would be 6.35 L, twice the pore volume [E1]. At 25 wt % the pores fill at 71 % RH at 20 °C, against 51 % RH at the v0.1 loading of 32.5 wt % (81 % RH at 20 wt %) [E2].

Kinetics limit what one humid night can do. A 10 h night at 90 % RH with the fan running, from the normal dusk state, takes the bed to 47 % of its pore volume [E3]. Three such nights in a row with overcast days between them, the worst case for a wet spell, fill 47, 57 and 66 % of the pores with the fan left running [E4], and 42, 48 and 54 % with the fan stopped above 70 % RH [E5]. No brine leaves the beads in either case, and the four drip-screen sumps (0.58 L) remain as a backstop [E6]. PR-007 (R7) is met on paper; a chamber test at TRL 4 must confirm it. The protection is kinetic, not equilibrium: a fog spell longer than three nights would keep filling the pores, and the 70 % RH cut-out sits 1 point below the 71 % pore-filling humidity, inside the ±2 to 3 % tolerance of a low-cost RH sensor (independent review; a lower cut-out is proposed to Amish in DWD-DEC-001).

## F. Electrical energy

The fan uses 20 Wh a night and the logger 3.6 Wh a day, 23.6 Wh in all, against 35 Wh a day from the 10 W panel, a margin of 1.48 [F1]. The 76.8 Wh battery, 80 % usable, gives 2.6 days with no sun [F2]. The fan works against only about 3.3 Pa: the 800 x 40 mm inlet above the deck, the 140 x 44 mm outlet slot behind the fan hood (DWD-DDR-003) and 0.16 Pa for the bed, so a 120 mm fan delivers 40 m³/h at low speed [F3]. The humidity cut-out only ever lowers the fan energy.

## G. Mass

v0.4 takes each component of the constructable model (DWD-DDR-003) with its own material: plywood skins at 0.60 g/cm³, softwood battens at 0.45, PIR at 0.035, aluminium at 2.70 and steel at 7.85, the drip-screen channels at the thickness of 0.5 mm flashing, and catalogue masses for the bought items drawn as blocks (bottle, fan, electronics box, PV panel, sensor shield). The box with its deck, trays, absorber plates, sorbent and drip screens weighs 36.1 kg, the stand 16.6 kg and the whole unit 61.1 kg (135 lb) dry; the walls, with their battens, are 7.6 kg [G1]. The trays with their absorbers, the sorbent, the deck and the drip screens (12.7 kg) all lift out through the open lid, and the box without them is 23.4 kg [G2]. v0.8 adds the four absorber plates (1.71 kg, perforated aluminium at 77.3 % solid) and replaces the perforated floor sheets with floor bars and a lighter glass-fibre mesh. v0.3 gave 33.4 kg for the box and 53.9 kg in total, from massing volumes with averaged densities.

## H. Wind

The design wind is a 25 m/s gust until the site value is known from the national wind annex (decided by Amish on 2026-10-25, DWD-DEC-001 item 4; it was 20 m/s with no gust allowance). At 25 m/s the dynamic pressure is 375 Pa and the normal force on the lid 495 N: 465 N of uplift and 169 N horizontal, against a dry weight of 600 N [H1]. The overturning moment of 424 N·m exceeds the restoring moment of 248 N·m (ratio 0.59), so the unit would tip without restraint; about 96 kg of ballast on the stand gives a factor of 1.5 [H2]. Without ballast or anchors it would also slide (margin -102 N); 38 kg of ballast gives 1.5 against sliding, and the anchor straps resist it [H3]. Two upwind ground anchors, each rated for 234 N or more in uplift, do the same job [H4]; they are the default restraint (DWD-DDR-002, D13) and are specified at 250 N or more each in the site soil (DWD-DEC-001, D20). For reference, at the earlier 20 m/s the lid saw 317 N and each anchor 96 N, or 39 kg of ballast [H5].

## I. Logging

Five channels every 5 min for 30 days are 8,640 records, about 0.41 MB as CSV [I1]; any microSD card holds years.

## J. Cost

The BOM has 16 lines and totals $695 against the $520 value-engineering target (`budget_usd`, a hypothetical control target, not a limit) [J1], so PR-011 (R11) is over the target by $175. The STEM check fixes of v0.8 added $56: the absorber plates ($24, line 4), black PTFE coatings and a PTFE-coated glass-fibre mesh ($32, lines 4 and 15), paint on the condenser underside ($8, line 6), and floor bars in place of the perforated floor sheets ($8 less, line 4), all estimates to quote. The v0.3 total was $515. The difference is the cost of making the design buildable (DWD-DDR-003): new line 16, the tray deck and wall ledges ($35); wall battens, inserts and paint ($13 more on line 2); the lid's U-channel frame, hinges and latches priced in full ($15 more); trays with a perforated floor under the mesh ($10 more in all); fin feet, rivets and epoxy ($4); a proper drain fitting ($7); stand bracing, foot plates and straps ($3); screen frames ($4 in all); pole, bracket and backing plate ($6); the fan hood ($1); and fixings ($7). The 2026-10-02 decisions added $19: a glazing sheet rated 130 °C or more, estimated at about $48 per square metre against $32 for a standard sheet ($17 more on line 3), and the printed "Empty before lifting" label ($2 on line 14). Cost drivers and savings worth trying are in the Value engineering section of the design decisions register. Over 5 years at the design point the parts cost about $0.87 per litre of water [J2].

## K. Requirements

*Table 4. Requirement status (DWD-PRS-001 v1.0; the old DWD-REQ-001 number in brackets). Also written to* `docs/04-calcs/results.csv`.

| ID | Requirement | Value | Target | Status |
|---|---|---|---|---|
| PR-001 (R1) | Water at the design point | 0.44 L/day; 0.37 to 0.41 under the sensitivities (C9, C12, C13) | 0.5 L/day or more | **Not met** |
| PR-002 (R2) | Water in dry nights | 0.31 L/day at 25 % RH | 0.25 L/day or more at 25 % RH | Met |
| PR-003 (R3) | Sun-only regeneration | Bed 105 °C at noon | 75 °C or more by midday | Met |
| PR-004 (R4) | Own electrical loads | 23.6 Wh/day against 35 Wh/day of PV; 2.6 days of autonomy | 30 Wh/day or less; 2 days or more | Met |
| PR-005 (R5) | Condenser rise | 11.9 K; about 14 to 16 K if the fins shed less heat than the correlation says | 15 K or less | Met on the correlation; at risk |
| PR-006 (R6) | Water quality | Food-grade materials by selection; chloride carry-over needs a water test | Food-grade; chloride below 250 mg/L | Not verifiable at TRL 3 |
| PR-007 (R7) | Salt containment | Pores fill at 71 % RH; 47 % of the pore volume after one 90 % RH night; 54 % after three with the fan rule; sumps 0.58 L | No brine leaves the tray and drip-screen assembly after a 90 % RH night | Met |
| PR-008 (R8) | Two actions per day | Open the flaps at dusk, close them at dawn; the fan runs on the logger timer with a humidity cut-out | 2 actions, 5 min or less | Met |
| PR-009 (R9) | Portable | Box 36.1 kg with everything inside; 23.4 kg with the trays and deck lifted out; total 61.1 kg | Box 35 kg or less when lifted with the trays and deck taken out (restated 2026-10-02, DWD-DEC-001 item 2); stand separable | Met |
| PR-010 (R10) | Survive the site | Two anchors of 234 N each (specified 250 N, default) or 96 kg of ballast for a 25 m/s gust (H2 to H4); UV life and 300 cycles need supplier data and test | Stable in a 25 m/s gust until the site value is known; UV-stable; 300 cycles | Not verifiable at TRL 3 |
| PR-011 (R11) | Cost | $695 | $520 or less (value-engineering target) | **Over the value-engineering target by $175** |
| PR-012 (R12) | Record performance | 8,640 records, 0.41 MB | Every 5 min for 30 days | Met |
| PR-013 (R13) | Protect users | Glazing outer skin about 55 °C at stagnation; fins up to 47 °C; 12.8 V DC | Touched surfaces 60 °C or less; below 60 V DC | Met |
| PR-014 | Local build and repair | Every made part needs only hand tools, a drill, a rivet tool, a folding bar or small brake and a coating shop (DWD-BLD-001, section 7) | No part needs special tooling | Met by design (not computed by the script) |

Counts from the script [K1]: 9 met, 1 not met (PR-001), 1 over its value-engineering target (PR-011), 2 not verifiable at TRL 3 (PR-006, PR-010); PR-014 is met by design.

## L. Numbers changed from v0.1

*Table 5. v0.1 figures against this revision. DWD-PRC-001 and DWD-REQ-001 v0.4 carry the new values.*

| Quantity | v0.1 (as drawn) | v0.2 (design) | Tag |
|---|---|---|---|
| Salt loading; bed depth | 32.5 wt %; 8.3 mm | 25 wt %; 9.3 mm | A5 |
| Overnight uptake | 0.23 kg | 0.60 kg | B5 |
| Water at 40 % night RH | 0.20 L per day | 0.57 L per day | C3 |
| Water at 25 % night RH | 0.09 L per day | 0.45 L per day | C4 |
| Bed at noon | 104 °C | 112 °C | C2 |
| Condenser rise; load | 13.1 K; 328 W | 12.1 K; 299 W | C2 |
| Solar-to-water efficiency | 2.5 % | 7.1 % | C7 |
| Stagnation, dry bed | 105 °C | 122 °C | D1 |
| Glazing outer skin; fins | 53 °C; 48 °C | 57 °C; 47 °C | D2, D3 |
| Pore-filling humidity | 51 % RH | 71 % RH | E2 |
| Mass; box | 51.4 kg; 30.9 kg | 53.9 kg; 33.4 kg | G1 |
| Ballast; anchor rating | 49 kg; 120 N | 46 kg; 114 N | H2, H4 |
| Parts cost | $488 | $515 | J1 |
| Water cost over 5 years | $1.34 per litre | $0.49 per litre | J2 |

*Table 6. Numbers changed by the constructable design (DWD-DDR-003). Water, heat, salt and electrical figures are unchanged.*

| Quantity | v0.3 | v0.4 | Tag |
|---|---|---|---|
| Box; total mass | 33.4 kg; 53.9 kg | 35.2 kg; 60.2 kg | G1 |
| Box with trays (and now the deck) out | 24.0 kg | 23.4 kg | G2 |
| Fan duty | about 0.4 Pa | about 3.3 Pa | F3 |
| Ballast; anchor rating | 46 kg; 114 N | 40 kg; 98 N | H2, H4 |
| Parts cost | $515 | $620 | J1 |
| Water cost over 5 years | $0.49 per litre | $0.59 per litre | J2 |

*Table 7. Numbers changed by the decisions of 2026-10-02 (DWD-DEC-001). Mass, water, heat and wind figures are unchanged.*

| Quantity | v0.6 | v0.7 | Tag |
|---|---|---|---|
| Parts cost; over the $520 target | $620; $100 | $639; $119 | J1 |
| Water cost over 5 years | $0.59 per litre | $0.61 per litre | J2 |
| Computed count for R9 | not met (first wording) | met (restated) | K1 |

*Table 8. Numbers changed by the STEM check fixes and the independent review of 2026-10-25 (DWD-DEC-001 D16 to D20). Salt-solution, electrical and logging figures are unchanged.*

| Quantity | v0.7 | v0.8 | Tag |
|---|---|---|---|
| Transmittance-absorptance | 0.74 (on white beads, wrongly taken as black) | 0.70 (black absorber plates) | C11 |
| Open share of the tray floor for the vapour | 0.60 (taken; about 0.25 in fact) | 0.59 (mesh on bars, D19) | C5 |
| Overnight uptake at 40 % RH | 0.60 kg | 0.47 kg | B5 |
| Water at 40 % night RH | 0.57 L per day | 0.44 L per day | C3 |
| Water at 25 % night RH | 0.45 L per day | 0.31 L per day | C4 |
| Bed at noon | 112 °C | 105 °C | C2 |
| Condenser rise; load | 12.1 K; 299 W | 11.9 K; 292 W | C2 |
| Solar-to-water efficiency | 7.1 % | 5.4 % | C7 |
| Stagnation, dry bed (design point; peak day) | 122 °C | 111 °C; 133 °C with the glazing inner skin at 106 °C | D1, D6 |
| Glazing outer skin; fins | 57 °C; 47 °C | 55 °C; 47 °C | D2, D3 |
| Box with everything inside; total | 35.2 kg; 60.2 kg | 36.1 kg; 61.1 kg | G1 |
| Ballast; anchor rating | 40 kg; 98 N | 96 kg; 234 N at the 25 m/s design gust (specified 250 N); 39 kg and 96 N at the earlier 20 m/s | H2, H4, H5 |
| Parts cost; over the $520 target | $639; $119 | $695; $175 | J1 |
| Water cost over 5 years | $0.61 per litre | $0.87 per litre | J2 |

## M. Limits of this note

- The isotherm is built from bulk salt thermodynamics. Salt confined in silica pores behaves differently, and the Conde parameters are used outside the range of the fit at the high-temperature, high-concentration end. Measured isotherms of the actual composite at 20 °C and at 80 to 120 °C must replace it. The Conde parameters should also be checked against the paper.

- The day model treats the vapour gap as still air and the drip screen as a simple added path and radiation shield. Any convection in the tilted box would speed up desorption; leaks at the flaps would lose vapour. Both need a test.
- Diffusion inside the grains is not modelled; C9 shows the yield falls to 0.38 L per day if it cuts the night NTU to 1.
- The bed is lumped at one temperature although the heat now enters at the absorber plate on top; C12 puts the cost at about 0.03 L per day. The share of light through the absorber's holes that is absorbed (0.70), the glazing transmittance (0.78 at normal incidence, no angle or soiling loss) and the mesh's open share (0.60) are estimates.
- The day is a clear sine with a constant 20 °C night; cloudy days and cooler or warmer nights are not modelled. U_TOP is a fixed 3.2 W/(m²·K), slightly optimistic above 100 °C.

## N. Independent review (2026-10-25)

A separate Claude session, given only this note, `sizing.py`, the STEM check DWD-STM-001 and the product requirements DWD-PRS-001, and none of the author's reasoning, tried to refute the headline results. It rebuilt the day model with stubbed geometry and reproduced the script (0.50 against 0.51 L per day, 108 against 109 °C at noon, 11.5 K) for the version it saw. What it found and what changed:

*Table 9. Independent review findings.*

| Item | Finding | Change |
|---|---|---|
| Tray floor open share | Refuted: 60 % of the bed area was taken as open to the vapour, but a 50 % perforated sheet under a fine mesh passes about 25 to 30 %; with 0.3 the yield falls to about 0.46 L per day (0.36 L in the script with 0.25) | D19: open floor of a coarser mesh on thin bars, 59 % open; separate open shares for vapour and radiation; yield 0.44 L per day |
| Yield against PR-001 | Weakened: the 2 % margin rested on estimates and the sensitivities were never combined | C13 added (0.37 L per day); PR-001 now **not met** on paper; recovery options to Amish (DWD-DEC-001) |
| Absorber | Arithmetic upheld; 0.70 for light through the holes is unmeasured and may be lower; no angle or soiling loss on the glazing | Kept as an estimate; listed in the limits; C13 uses 0.5 |
| Stagnation | Weakened: run at 785 W/m², and the bed rather than the glazing skin was compared with the rating | D6 added: 133 °C bed, 106 °C inner skin at 1000 W/m² and 40 °C; S13 now checks the skin; S27 added for the other hot materials |
| Wind | Weakened: 94 N against a 100 N anchor is a 6 % margin; gusts not checked | H5 added; anchors specified at 250 N (D20); design wind speed to Amish |
| Condenser | Weakened: the isolated-plate correlation is optimistic for fins under a hot tilted floor; PR-005 fails at about 30 % less convection and half the radiation | PR-005 marked at risk; S11 plausible |
| Salt containment | Weakened: protection is kinetic; the 70 % RH cut-out is 1 point below pore filling, inside sensor tolerance | Lower cut-out proposed to Amish (decided value of DWD-DDR-002, D11); decided 2026-10-25: 65 % RH, restart below 60 % |
| Efficiency | Weakened: 6.3 % is high for a single stage | S8 plausible; now 5.4 % |
| Energy balance, uptake, vapour pressure, touch temperatures, coatings | Upheld (S10's value and limit labels clarified) | None |
| - The natural airflow with the fan stopped (4 m³/h) is an assumption; E4 shows that R7 holds even if the fan keeps running. | | |

**Safety:** A proof of concept on paper does not show that the water is safe to drink. R6 (water quality) needs a laboratory water test before anyone drinks the water, and the documents keep the instruction to test and treat it.